Surrogates for Monte Carlo uncertainty propagation with twenty model runs: a controlled comparison

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Abstract

Monte Carlo propagation of distributions (GUM Supplement 1) needs 10^5–10^6 evaluations of the measurement model. When the model is an expensive simulation, it is replaced by a surrogate trained on a handful of runs, and the surrogate's own error enters the uncertainty budget. We compare surrogate strategies for a budget of twenty runs on four test models: a benchmark from an earlier analysis whose dominant term matches one candidate surrogate, a Gaussian plume model of an ammonia release, and the borehole and Ishigami functions. Strategies cover single Gaussian processes (GPs) with five kernel specifications and log-transformed variables, nine-member committees of different surrogates combined uniformly or by cross-validation (CV) error weights, and adaptive designs. Each is run with 30 paired seeds and scored on the propagated mean, 95th percentile and exceedance probabilities against reference Monte Carlo evaluations. Kernel specification changed the error up to eighteen-fold: an amplitude hyperparameter held fixed produced an apparent GP failure that the earlier analysis had attributed to domain size. Logarithmic transformation of positive variables reduced the error up to eightfold. A committee beat a well-specified GP only where one member matched the response structure, and only with CV weighting; on the borehole model it was 2.3 times worse. The adaptive designs tested gave no consistent gain, and a spectral acquisition criterion was never better than plain variance. Five-fold CV estimated the surrogate error conservatively, whereas the GP's own predictive standard deviation understated it by up to 2.8 times where the GP was weakest. We give practical rules for surrogate-based uncertainty evaluation and a conservative way to include the surrogate contribution.

1 Introduction

Evaluating measurement uncertainty by the propagation of distributions, following Supplement 1 to the Guide to the Expression of Uncertainty in Measurement (GUM S1) [1,2], requires the measurement model $Y = f(\mathbf{X})$ to be evaluated once per Monte Carlo trial, typically 10^5 to 10^6 times [3,4]. For many quantities of practical interest f is a numerical simulation. A computational fluid dynamics (CFD) model of an accidental ammonia release [5], for example, takes hours to days per run, and a practical budget is often fewer than twenty runs.

The usual remedy is a *surrogate* (emulator) trained on the available runs and used in place of f [6–8]. Its error is then a contribution to the uncertainty budget that must be evaluated, not assumed negligible; Kennedy and O'Hagan [9] treat such model inadequacy as an explicit uncertainty source. Gaussian process (GP) regression [10,11] is the standard surrogate for computer experiments. A widely used guideline asks for about ten runs per input [12], and GP results are known to depend on modelling choices that are easy to get wrong [13]. For smaller budgets, alternatives include weighted committees of different surrogates [14,15], adaptive selection of runs where the surrogate is uncertain [16], and, recently, spectral measures of disagreement [17].

An earlier analysis by the first author combined these ideas: a heterogeneous committee, a spectral acquisition criterion (UEigV) and dynamic model management. It reported that a passively sampled GP fails catastrophically on large input domains with a high output dynamic range, and that the committee recovers accuracy at the same budget. This paper re-examines those claims with correctly specified baselines and extends the comparison to models where no committee member matches the response. Its contributions are:

1. a controlled, paired comparison of single-GP, committee and adaptive strategies at a budget of twenty runs on four models, scored on the quantities a GUM S1 evaluation reports;
2. the finding that the reported GP failure is a kernel-specification artefact, and that kernel specification and variable transformation matter more than the choice of strategy;
3. a characterisation of when a committee helps (a member matching the dominant structure, combined by error weights) and when adaptive designs help (inconsistently, for the criteria tested);
4. an assessment of how far the surrogate's error can be estimated from the training runs alone, and a conservative way to include it in a GUM S1 evaluation.

2 Problem and protocol

2.1 Surrogate-based propagation

Given input distributions $g_{\mathbf{X}}$ and N runs $\{(\mathbf{x}_i, f(\mathbf{x}_i))\}$, a surrogate $\hat{f}$ is fitted and the Monte Carlo method of GUM S1 is applied to $\hat{f}$ instead of f. Here $N = 20$, and the propagated mean, standard deviation (SD), 95th percentile $y_{0.95}$ and exceedance probabilities are compared with those obtained from f itself. All input distributions are uniform over the ranges given below.

2.2 Test models

Physics is the benchmark of the earlier analysis, $f(\mathbf{x}) = \sin 2x_1 \cos 2x_2 + \mathrm{e}^{-x_3^2} + 0.2\|\mathbf{x}\|^2$ on $[-h, h]^3$ with $h = 200$

unless stated, observed with noise of SD 0.1. Its quadratic term carries 44% of the output variance at $h = 2$ and practically all of it for $h \geq 10$, and lies exactly in the class of a degree-2 polynomial chaos expansion.

Plume is the ground-level concentration of a continuous release from a Gaussian plume model with ground reflection,

$$C = \frac{10^3\, Q}{2\pi\, u\, \sigma_y \sigma_z} \exp\!\left(-\frac{y^2}{2\sigma_y^2}\right) 2\exp\!\left(-\frac{H^2}{2\sigma_z^2}\right), \qquad (1)$$

in $\mathrm{mg\,m^{-3}}$, with Briggs rural dispersion coefficients for neutral stability, $\sigma_y = 0.08x(1 + 10^{-4}x)^{-1/2}$ and $\sigma_z = 0.06x(1 + 1.5 \times 10^{-3}x)^{-1/2}$. Inputs: emission rate $Q \in [100, 2000]\,\mathrm{g\,s^{-1}}$, wind speed $u \in [1, 6]\,\mathrm{m\,s^{-1}}$, release height $H \in [1, 20]$ m, downwind distance $x \in [200, 1000]$ m and crosswind offset $y \in [-30, 30]$ m. The output spans 1.8 to 2400 $\mathrm{mg\,m^{-3}}$, and the probability of exceeding the level immediately dangerous to life or health (IDLH) for ammonia, 210 $\mathrm{mg\,m^{-3}}$ (about 300 ppm), is 0.123. The plume model is an analytical proxy for dispersion problems, cheap enough for exact reference values; it is not a CFD simulation.

Borehole (water flow through a borehole, eight inputs, standard ranges) and *Ishigami* ($\sin x_1 + 7\sin^2 x_2 + 0.1x_3^4 \sin x_1$ on $[-\pi, \pi]^3$) are standard test functions for computer experiments. No candidate surrogate matches the plume, borehole or Ishigami response.

2.3 Strategies

Single GPs. Each is fitted to a 20-point Latin hypercube. Five kernel specifications are compared: (i) the original kernel of the earlier analysis, a squared exponential with one length scale and an amplitude that is *fixed* at 1 on the standardised output rather than estimated; (ii) with the amplitude estimated; (iii) with one length scale per input (automatic relevance determination, ARD) but the amplitude fixed; (iv) with both; (v) a Matérn-5/2 ARD kernel with amplitude. Hyperparameters are estimated by maximising the log marginal likelihood (L-BFGS-B, five random restarts) on standardised inputs and outputs, with length scales in $[10^{-2}, 10^3]$, amplitude in $[10^{-3}, 10^3]$ and a nugget variance in $[10^{-8}, 10^{-1}]$. A sixth variant fits (v) to $\log Y$, with logarithmic transformation of inputs whose range is strictly positive (plume and borehole); predictions are back-transformed as exp of the latent mean, i.e. the predictive median, without bias correction. The adaptive GP starts from five Latin hypercube runs and adds 15 at its largest posterior SD over a random candidate pool [16].

Committees. Nine members: two GPs (squared exponential and Matérn-3/2, with amplitude and ARD), polynomial chaos expansions (PCE) [18,19] of degrees 2 and 3 (Bayesian ridge regression), random forests of 50 and 100 trees, a support vector regressor and two neural networks (two hidden layers of 64 and 128 units). Members are combined uniformly or with weights

$$w_k \propto \left[\mathrm{MSE}_{\mathrm{CV},k}\,(1 + \pi_k)\right]^{-1}, \qquad (2)$$

where $\mathrm{MSE}_{\mathrm{CV},k}$ is the three-fold cross-validation error of member k (a single 80/20 hold-out for the networks) and $\pi_k \geq 0$ penalises predictions below a known physical bound [14, 15]. A committee with the original GP kernel is included for reference. The passive committee is fitted to the same 20-point design as the single GPs. Adaptive committees start from five runs and add 15 where the active members disagree most, scored by the variance of their predictions or by UEigV.

UEigV. Lin et al. [17] proposed this measure for black-box language models, from the normalised graph Laplacian of pairwise similarities between sampled outputs. Here the graph is built from the K standardised member predictions $\hat{y}_k$ at a candidate input, with $A_{ij} = \exp(-(\hat{y}_i - \hat{y}_j)^2/2)$, $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$ and $\mathrm{UEigV} = \sum_k \max(0, 1 - \lambda_k)$, a continuous count of agreement clusters.

2.4 Evaluation

Each strategy is run with seeds 1000–1029. Seeds are shared and the Latin hypercube is seeded from the global random state, so passive strategies use the identical design and adaptive ones the identical start. Accuracy over the input domain is the root mean square error (RMSE) on 5000 test points divided by the output SD (normalised RMSE; 1 corresponds to a constant predictor). The surrogate propagates 3×10^4 input draws; the reference uses 2×10^5 draws of f. Two exceedance assessments are distinguished: $P(Y > y_{0.95})$ at the reference 95th percentile, whose true value is 0.05 by construction, tests tail accuracy on every model; $P(C > \mathrm{IDLH})$ at a fixed physical threshold is the quantity a risk decision uses. Strategies are compared by two-sided Wilcoxon signed-rank tests on paired log normalised RMSE. The 36 tests are treated as exploratory and their p-values are Holm-adjusted. All errors above are *validation* errors, which need the reference model; Section 3.4 examines what can be estimated without it.

3 Results

3.1 Kernel specification and variable transformation

Table 1 lists every strategy on every model and Table 2 the paired comparisons. On the physics model the original kernel reaches a normalised RMSE of 0.59; estimating the amplitude alone reduces it to 0.033, an eighteen-fold improvement in all 30 seeds (adjusted $p < 10^{-6}$). ARD alone does not help (0.61). With the amplitude fixed at the sample variance of the standardised output, the GP cannot represent a quadratic bowl whose values far exceed one SD and reverts towards the mean between the runs.

Fig. 1 repeats this over the domain sweep of the earlier analysis. With the original kernel the normalised RMSE is 0.60 at every half-width from 10 to 200, which the earlier analysis read as a failure of GPs on large domains. A Matérn-5/2 ARD GP with amplitude reaches 0.17 at $h = 10$ and 0.08 from 50 to 200. The committee of the earlier analysis remains more accurate, falling to 1.7×10^{-4} at $h = 200$, but because its degree-2 PCE represents the quadratic term exactly: its median RMSE stays at 0.62–0.68, close to the RMS of the oscillatory and exponential terms (0.50–0.61) that no member can learn from twenty runs.

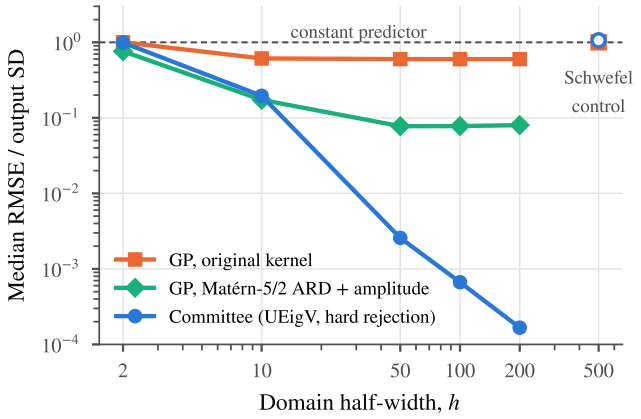


Figure 1: Physics model on $[-h, h]^3$: median normalised RMSE over 30 seeds against the domain half-width, for a GP with the original kernel (amplitude fixed), a Matérn-5/2 ARD GP with amplitude, and the committee of the earlier analysis. Open markers: Schwefel function on $[-500, 500]^3$. Dashed line: constant predictor.

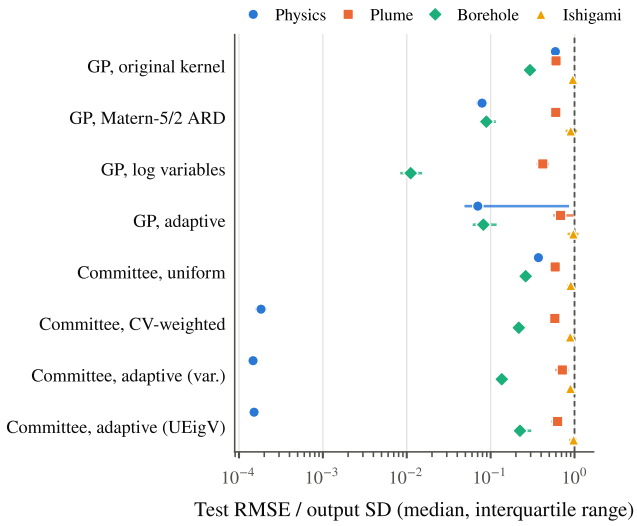


Figure 2: Normalised RMSE of each strategy on each model: median (marker) and interquartile range (bar) over 30 seeds. Dashed line: constant predictor.

On the borehole model ARD is what matters (0.30 to 0.09), and on the plume and Ishigami models no kernel specification helps. Logarithmic transformation gives the largest single improvement where it applies: 0.089 to 0.011 on the borehole model (eightfold, all 30 seeds) and 0.60 to 0.42 on the plume (27 of 30 seeds). No strategy does better than 0.89 on the Ishigami function, for which the ten-runs-per-input guideline [12] would ask for 30 runs.

3.2 Committees, aggregation and adaptive runs

Fig. 2 summarises all strategies. A committee outperforms the best single GP only on the physics model, by more than two orders of magnitude, and only through its degree-2 PCE, on which the CV weights concentrate. On the borehole model the CV-weighted committee is 2.3 times worse than the Matérn-5/2 GP on the identical design (0 of 30 seeds better, adjusted $p < 10^{-6}$); on the plume and Ishigami models the two tie.

Aggregation matters wherever a committee helps. Uniform averaging is worse than CV weighting on the physics model by a factor of about 2000 and on the borehole model by about 20% (adjusted $p \leq 0.002$), with either GP specification, and is never better. On the physics model a prior-reverted or otherwise poor member is enough to move a uniform mean by orders of magnitude even when an exact member is present.

Within the designs tested (five initial runs; 15 adaptive runs chosen by GP posterior SD, committee variance or UEigV over a random candidate pool), adaptive selection gave no consistent benefit. For the single GP it was never significantly better and was worse on the plume before adjustment. For the committee it helped on the physics and borehole models (median ratios 0.80 and 0.61), hurt on the plume (1.25, adjusted $p = 0.005$; median error of the propagated mean 55%) and made no difference on the Ishigami function. UEigV was never better than variance acquisition after adjustment and was 1.7 times worse on the borehole model (adjusted $p < 10^{-6}$). Criteria aimed at a particular quantile or threshold [16] may behave differently.

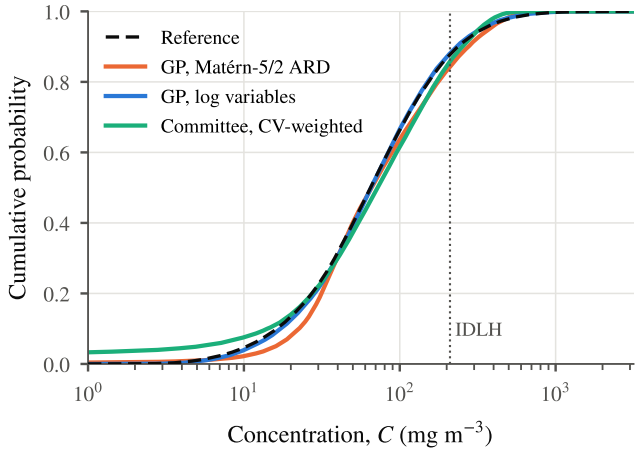


Figure 3: Plume model, first seed: distribution of the ground-level concentration propagated through three surrogates fitted to the same 20 runs, and the reference Monte Carlo distribution. Dotted: IDLH level for ammonia.

3.3 Propagated quantities

Table 3 gives validation errors of the propagated quantities. They follow the normalised RMSE. With the best strategy for each model, the 95th percentile is within 0.41% (physics), 6.9% (plume), 0.60% (borehole) and 16% (Ishigami) of the reference, and $P(Y > y_{0.95})$ within 0.0015, 0.0076, 0.0018 and 0.029 of 0.05. At the fixed IDLH threshold (reference $P(C > \text{IDLH}) = 0.123$), the log-variable GP has a median absolute error of 0.009 over 30 seeds against 0.023 for the raw-variable GP; Fig. 3 shows the first seed, for which the log-variable GP gives 0.119, the raw-variable GP 0.158 and the CV-weighted committee 0.142.

3.4 Estimating the surrogate contribution

In practice the reference is unavailable and the surrogate's contribution must be estimated from the training runs. Ta-

Table 1: Normalised RMSE, median (interquartile range) over 30 seeds; $N = 20$ runs. Committees use GP members with amplitude and ARD unless marked "orig. GPs". Generated from the result records.

Strategy	Physics	Plume	Borehole	Ishigami
GP, original kernel (amp. fixed)	0.591 (0.56–0.61)	0.602 (0.55–0.63)	0.295 (0.27–0.33)	0.956 (0.91–1)
GP, + amplitude	0.033 (0.027–0.037)	0.59 (0.54–0.6)	0.247 (0.23–0.27)	0.97 (0.92–1)
GP, + ARD	0.608 (0.57–0.66)	0.632 (0.59–0.67)	0.129 (0.11–0.15)	1.01 (0.9–1.1)
GP, + amplitude + ARD	0.0361 (0.027–0.041)	0.596 (0.57–0.67)	0.0848 (0.074–0.1)	0.896 (0.81–1)
GP, Matérn-5/2, amp. + ARD	0.079 (0.069–0.087)	0.597 (0.56–0.64)	0.0889 (0.08–0.11)	0.902 (0.81–1)
GP, Matérn-5/2, log variables	–	0.418 (0.37–0.48)	0.0111 (0.0087–0.015)	–
GP, adaptive (posterior SD)	0.0703 (0.051–0.83)	0.681 (0.58–0.94)	0.0819 (0.063–0.11)	0.97 (0.86–1.1)
Committee (orig. GPs), uniform	0.469 (0.45–0.5)	0.608 (0.56–0.65)	0.317 (0.3–0.36)	0.957 (0.91–1)
Committee (orig. GPs), CV	0.000183 (0.00016–0.00021)	0.589 (0.56–0.64)	0.295 (0.27–0.33)	0.964 (0.91–1)
Committee, uniform	0.371 (0.36–0.41)	0.588 (0.56–0.63)	0.261 (0.25–0.3)	0.903 (0.85–0.96)
Committee, CV	0.000183 (0.00016–0.00021)	0.582 (0.53–0.62)	0.217 (0.19–0.26)	0.889 (0.84–0.97)
Committee, adaptive (variance)	0.000147 (0.00014–0.00016)	0.716 (0.61–0.82)	0.136 (0.12–0.16)	0.895 (0.87–0.93)
Committee, adaptive (UEigV)	0.000151 (0.00015–0.00017)	0.628 (0.55–0.69)	0.224 (0.21–0.29)	0.975 (0.9–1)

Table 2: Paired comparisons over 30 seeds: median ratio of normalised RMSE (first / second; below 1 favours the first), number of seeds where the first is better, and Holm-adjusted p (two-sided Wilcoxon signed-rank on log normalised RMSE, 36 tests). Generated from the result records.

Comparison	Physics	Plume	Borehole	Ishigami
GP Matérn ARD + amp. vs original	0.138; 28; $< 10^{-6}$	0.973; 18; 1.00	0.308; 30; $< 10^{-6}$	0.951; 18; 1.00
GP + amp. vs original	0.0554; 30; $< 10^{-6}$	0.968; 24; 0.03	0.829; 26; 6.6×10^{-6}	1; 12; 0.84
CV vs uniform (orig. GPs)	0.000381; 30; $< 10^{-6}$	0.975; 24; 2.9×10^{-3}	0.933; 26; 2.5×10^{-3}	1; 15; 1.00
CV vs uniform	0.000473; 30; $< 10^{-6}$	0.983; 23; 0.28	0.844; 25; 1.6×10^{-3}	0.991; 18; 1.00
Committee (CV) vs GP Matérn	0.00225; 29; $< 10^{-6}$	0.988; 16; 1.00	2.33; 0; $< 10^{-6}$	0.988; 16; 1.00
Adaptive vs passive GP	1.01; 15; 1.00	1.18; 11; 0.12	0.945; 15; 1.00	1.06; 12; 1.00
Adaptive vs passive committee	0.796; 26; 4.0×10^{-5}	1.25; 6; 4.6×10^{-3}	0.61; 26; 1.7×10^{-5}	0.981; 16; 1.00
UEigV vs variance acquisition	1.02; 10; 0.27	0.9; 20; 0.12	1.67; 1; $< 10^{-6}$	1.07; 9; 0.07
GP log variables vs raw	–	0.691; 27; 3.7×10^{-4}	0.109; 30; $< 10^{-6}$	–
GP log variables vs committee	–	0.735; 24; 1.9×10^{-3}	0.0541; 30; $< 10^{-6}$	–

ble 4 compares, for the recommended GPs, the true error with two estimates available in practice: the five-fold CV RMSE with hyperparameters re-fitted in every fold, and the root mean square of the GP's own predictive SD over the domain.

CV was conservative. It was at least the true error in 28 and 30 of 30 seeds on the physics and borehole models, where the GP is accurate and the held-out folds are relatively harder, close to it on the plume and Ishigami models, and below half the true error in at most 6 of 30 seeds (plume, log variables). The GP's own uncertainty was unreliable in the unsafe direction where the GP is weakest: on the plume and Ishigami models it understated the error 1.7–2.8 times and the nominal 95% predictive intervals covered 73–77% of test points; on the physics model and the log-variable borehole GP it overstated the error. The member envelope of a committee, read as an interval for $P(Y > y_{0.95})$ [20], contained the reference value in 0 to 100% of seeds depending on model and strategy, and is not a substitute.

These results suggest a simple, conservative treatment within GUM S1: propagate $Y = \hat{f}(\mathbf{X}) + E_s$, where the surrogate error E_s is an additional input with expectation zero and standard uncertainty equal to the CV RMSE, independent of $\mathbf{X}$. It ignores the spatial variation and correlation of the surrogate error and inflates the uncertainty where the surrogate is accurate. Calibrated alternatives, for example conformal prediction [21], need validation on a CFD case.

4 Relation to the earlier analysis

The earlier analysis made five claims. Three are withdrawn or revised. The catastrophic GP failure on large domains is a kernel-specification artefact (Section 3.1). The spectral acquisition criterion gives no advantage over plain committee variance; on the physics model the earlier paired campaign already favoured variance ($p = 0.047$, 30 versus 27 of 30 successes), and here UEigV is never better. The committee's advantage is not general but comes from a member matching the dominant term. Two claims survive and are strengthened: aggregation must be error-weighted, and hard rejection of members is a degenerate baseline (in the earlier campaign it collapsed to one member in all 30 trials and then sampled at random for 31% of the budget). The records of those campaigns are kept in the repository [22].

5 Discussion and limitations

The study is deliberately narrow: four analytical models, uniform inputs, one budget ($N = 20$), one design size for adaptive starts, and acquisition by variance-type criteria. The plume model represents the structure of a dispersion problem but not the irregularities of a CFD solution, such as recirculation, regime changes or numerical noise; a CFD case is the necessary next step. Results at larger budgets may differ, in particular for adaptive designs. The paired tests are

Table 3: Validation error of propagated quantities, median over 30 seeds: relative error (%) of the mean and of $y_{0.95}$, and absolute error ($\times 100$) of $P(Y > y_{0.95}) = 0.05$. Bold: best per column. Generated from the result records.

	Physics			Plume			Borehole			Ishigami		
Strategy	mean	$y_{0.95}$	P	mean	$y_{0.95}$	P	mean	$y_{0.95}$	P	mean	$y_{0.95}$	P
GP, original kernel	8.90	21.1	4.86	10.4	20.4	3.29	1.60	13.0	4.25	10.3	34.6	4.73
GP, Matérn-5/2 ARD	1.36	3.88	1.37	9.68	17.7	2.87	0.66	2.45	0.91	**8.48**	**16.0**	3.61
GP, log variables	–	–	–	**4.74**	**6.94**	**0.76**	**0.17**	**0.60**	**0.18**	–	–	–
Committee, CV-weighted	**0.21**	**0.41**	**0.15**	8.75	23.1	3.15	1.06	10.9	3.05	11.0	28.7	**2.89**
Committee, adaptive	**0.21**	**0.41**	**0.15**	55.1	20.5	5.39	3.30	1.41	0.49	15.9	34.9	4.40

Table 4: Estimating the surrogate error from 20 runs, median over 30 seeds: ratio of the five-fold CV RMSE and of the GP's root-mean-square predictive SD to the true RMSE, and coverage of the nominal 95% predictive interval. M: Matérn-5/2 ARD GP with amplitude; L: the same on log variables.

Model	GP	CV/true	SD/true	Coverage
Physics	M	9.06	1.58	100%
Plume	M	0.99	0.36	77%
Plume	L	0.81	0.42	73%
Borehole	M	2.11	0.79	90%
Borehole	L	2.61	1.45	99%
Ishigami	M	1.07	0.58	73%

exploratory and Holm-adjusted; several nominal differences (uniform versus CV weighting on the plume, adaptive versus passive GP on the plume, UEigV versus variance on three models) do not survive the adjustment and are not claimed. The log transformation applies only to positive outputs, and its back-transformation to the predictive median slightly underestimates the mean of a skewed output. Finally, the committee's components and hyperparameters were not tuned per model, which reflects practice at this budget but may understate what a tuned committee could achieve.

6 Practical rules

For surrogate-based GUM S1 evaluations with about twenty runs:

1. *Specify the GP fully*: estimate an amplitude and one length scale per input; inspect the fitted hyperparameters.
2. *Transform the variables* when the model is multiplicative or positive, for example concentrations and flows.
3. *Use a committee only with error weighting*, and expect a gain only when a member can represent the dominant structure.
4. *Prefer a space-filling design* unless an adaptive criterion has been shown to help for the quantity of interest.
5. *Estimate the surrogate contribution by cross-validation* and include it in the budget; do not rely on the GP's own predictive variance or on committee spread.

7 Conclusion

For Monte Carlo uncertainty propagation with twenty runs of an expensive model, the kernel specification of a GP surrogate and the transformation of the variables mattered more than the choice between single surrogates, committees and the adaptive designs tested. A GP failure reported in an earlier analysis was caused by one hyperparameter held fixed. Committees helped only when a member matched the response structure and only with CV error weighting. Cross-validation gave a conservative estimate of the surrogate's error; the GP's own uncertainty did not. A CFD dispersion case, other budgets and a calibrated surrogate contribution are the next steps.

Data and code availability

All code, per-seed records, figure scripts and an audit script that recomputes every reported number are available at `https://github.com/Yugam2508/fyp-active-learning-cfd` [22].

Authors' contribution

Yugam Jinesh Vora: conceptualisation, methodology, software, investigation, formal analysis, visualisation, writing – original draft. Wee Hoe Ng: conceptualisation, supervision, writing – review and editing.

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